



Validation of the use of concept maps as an evaluation tool for the teaching and learning of mechanical and industrial engineering

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2024 © The Author(s) 2024 **Abstract**

This paper presents the experimental work developed to measure the learning process through concept map analysis. The development of a concept map is requested by the students for each chapter or theme of the subject. As a result, maps from engineering courses have been analyzed. The measurements carried out consider several parameters, such as individual and team map building, student progressive knowledge level, and map complexity. Concerning the complexity analysis, the focus is qualitative, and it is based on the data extracted from the concept maps elaborated by the students. The study, conducted during the 2018–2019 academic year, included students from various academic levels and institutions, such as the Public University of Navarra UPNA and the University of the Basque Country UPV-EHU, covering first-degree students of Bachelor's Degree in Mechanical Engineering and first-degree students of Master's Degree in Industrial Engineering at UPNA, third-degree students of Bachelor's Degree in Mechanical Engineering at UPV-EHU. The data collected from 37 individual maps in Industrial Drawing, 31 group maps in Industrial Drawing, 12 individual maps in Design of Machinery, and 12 group maps in Design of Machinery, along with a control group of 79 students who did not participate in any activity, provided valuable insights into the effectiveness of concept maps for evaluating understanding levels and learning outcomes across various engineering subjects and academic levels. The learning outcome of the students is treated to obtain the level of understanding of complex systems shown by the students through the concept maps previously drawn and the questionnaire answered by each student about the achievement of learning results through the use of concept maps. This work shows the research methodology established and the learning results achieved qualitatively: measuring the maps by means of a rubric, self-assessment based on a survey, and through the questionnaires. Also, the results obtained in the final exams have been compared. From the observed results, this methodology is presented as a suitable alternative for evaluating the correct acquisition of concepts in online teaching situations.

Keywords Concept maps · Assessment tool · Complexity · Teaching and learning

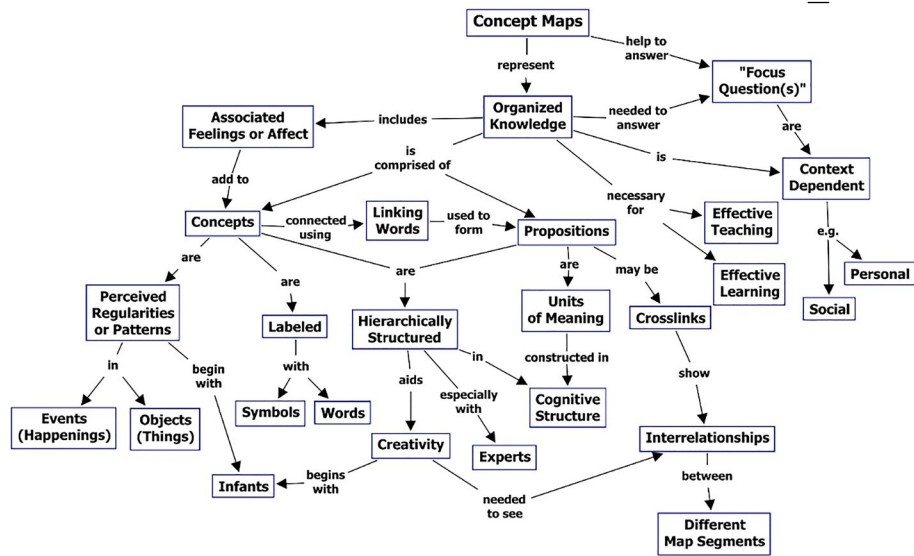


Fig. 1 Concept map: set up of elements and links (Cañas et al., 2005)

Introduction

Significant learning is learning with a meaning, referring to the use of the student's previous knowledge to construct new learning, which has become very relevant in recent years (Ausubel, 1963). One of the tools to achieve meaningful learning are concept maps, both student-oriented (Ausubel, 1963; Novak & Gowin, 1984), which support the organisation and representation of students' knowledge by ordering ideas and concepts (Novak, 1987; Novak et al., 1983). One of the main advantages of this tool is that it facilitates the transmission of information between students and teachers. It helps especially in communication in the bottom-up direction, from the student to the teacher.

Organising and representing knowledge is the main objective of concept mapping. This is done by means of graphical tools. Some of the elements included are: i) concepts usually enclosed in circles or squares; ii) relationships between concepts by means of connecting arrows; and iii) linking words or linking phrases placed above the arrows specifying the relationship between concepts, as can be seen in Fig. 1.

These concepts, arrows, and linking words are used in the visual representation in a hierarchical way, arranged in the form that each person has in his or her mind. When we speak and/or read, this hierarchy becomes linear. In order for someone to achieve meaningful learning with the information collected in a linear mode, it has to be given a hierarchical structure. Concept maps work as "translators" to pass hierarchically arranged information into linear mode and vice versa.

Concept maps have been used since the early 1990s. (Roth & Roychoudhury, 1994) showed their usefulness as devices to collect and reflect on the concepts acquired in collaboration between students and teachers. The analysis of these conceptual maps allows for the diagnosis of conceptual errors in the architecture of the knowledge acquired by the student (Hwang, 2003; Panjaburee et al., 2010). Its use has also been extended to assess student learning (Markham et al., 1994; Schmid & Telaro, 1990). Recent research

highlights their positive impact on student perceptions and academic outcomes (Lenski et al., 2022). This study investigated the effects of concept map construction (CM-c, i.e., creating concept maps) and concept map study (CM-s, i.e., observing concept maps) on learning performance, concept map quality, cognitive load, accuracy of self-evaluation, and enjoyment in junior high school students. The results showed that CM-c training increased learning performance and concept map quality, and these effects transferred to learning with CM-s. Conceptual mapping tools have been increasingly utilized for various educational purposes by academicians and educators in recent years. Furthermore, technological advancements, including conceptual mapping, have significantly influenced the sustainability of education, as has been pointed out by (Almulla & Alamri, 2021). However, students' understanding and motivation to use conceptual mapping in the context of sustainable education have seldom been evaluated. Therefore, this study shows the benefits of developing and testing theories related to the use of conceptual mapping, as well as for practitioners who employ conceptual mapping in sustainable education.

Concept maps have proven to be versatile and effective tools across various academic disciplines. In Mathematics, (Evans & Jeong, 2023) demonstrated their utility as an assessment tool in university courses, revealing a statistically significant correlation between concept mapping performance and final exam scores, as well as emotional regulation factors of assessment self-efficacy. Similarly, in Medicine, concept maps have been instrumental in aiding students' comprehension of complex patterns and relationships, serving both as instructional aids and assessment tools in medical education (Sy et al., 2016; Torre & Daley, 2013). Moreover, in Science and Business fields, concept maps have been employed for research, instructional structuring, and project planning, illustrating their adaptability and effectiveness across diverse educational contexts (Lenski et al., 2022; Maker & Zimmerman, 2020; Olukoya & Jimoh, 2022).

These examples underscore the profound impact of concept maps on enhancing learning outcomes and academic performance across various domains. As highlighted by (McClure et al., 1999), concept maps serve multiple functions in education, including facilitating learning, guiding instruction, aiding in study plan preparation, and evaluating understanding of scientific concepts. In summary, concept maps are recognized as an effective strategy for fostering meaningful learning experiences in higher education, positively influencing students' perceptions of the learning environment and their motivation to engage with course materials.

The disciplines of Mechanical and Industrial Engineering cover the principles of engineering, physics, mathematic and materials science and require an understanding of the basis of mechanics, dynamics, thermodynamics, etc. A meta-analysis published in (Jackson et al., 2023) investigated the overall effectiveness of conceptual mapping-based education experiments on students of engineering education, suggesting that concept mapping can be a powerful tool in education. Finally, while some researchers suggest that concept mapping can increase learner motivation, others assert that the effects of this technique might vary among learners of different abilities, indicating that the effectiveness of concept maps may depend on the individual learner's abilities and learning style (Alt et al., 2023). Due to the complexity level of these Engineering fields, in this study, a few research questions are proposed in order to approach the conceptual understanding of the students.

The research holds significant importance in the realm of education for Mechanical and Industrial Engineering by leveraging conceptual mapping as a powerful tool for enhancing

meaningful learning. In a discipline as intricate as Mechanical and Industrial Engineering, where foundational concepts span across engineering principles, physics, drawing, manufacturing, and materials science, the introduction of conceptual mapping serves as a transformative approach. By employing concept maps, the study seeks to delve into the depth of conceptual understanding among students, allowing for a nuanced exploration of the fundamental principles underpinning mechanics, manufacturing, designing, and other critical areas. This research aims to bridge the gap between the complexity of these engineering fields and the effective comprehension of students, offering a valuable method to diagnose conceptual errors, reflect on acquired knowledge, and assess learning outcomes.

The contribution of this research lies not only in its innovative application of conceptual mapping but also in addressing specific research questions tailored to the challenging nature of Mechanical and Industrial Engineering. Through the analysis of concept maps, the study not only serves as a diagnostic tool for conceptual errors but also as a means to evaluate the students' understanding of scientific concepts within these disciplines. As a result, the research not only pioneers the use of conceptual mapping in the context of Mechanical and Industrial Engineering education but also provides a methodological framework that can be adopted to enhance learning outcomes and instructional strategies in these complex engineering fields. Based on the observed results, this methodology is presented as a suitable alternative for evaluating the correct acquisition of concepts in online teaching situations.

Hypotheses

The main hypotheses raised by this study are:

- The introduction of conceptual mapping as an instructional tool significantly enhances meaningful learning in Mechanical and Industrial Engineering education, supported by findings from studies such as (Evans & Jeong, 2023; Lenski et al., 2022), which demonstrate the efficacy of concept mapping in improving comprehension and retention of key concepts.
- Students utilizing conceptual mapping will demonstrate a higher level of comprehension and retention of essential concepts compared to traditional teaching methods, as evidenced by research conducted by (Jackson et al., 2023; Sy et al., 2016), indicating the superior effectiveness of concept mapping in facilitating learning outcomes.
- Conceptual mapping is well-received and accepted by students, particularly in online teaching environments, as suggested by (Alt et al., 2023), underscoring its efficacy as a pedagogical approach in complex engineering fields and aligning with the practical implications highlighted by (Olukoya & Jimoh, 2022) regarding its application in enhancing collaboration and understanding.

Materials and methods

The experimental activity of concept map testing was carried out in four student groups from two different universities (University 1 and University 2). Table 1 shows detailed information about the studied subjects and their academic specifications. In designing the

methodology for this study, the authors aimed to create a comprehensive and nuanced understanding of the effectiveness of conceptual mapping in various academic contexts. The choice of student groups from two different universities, University 1 and University 2, was deliberate. This decision was rooted in the intention to compare

Table 1 Experimentally studied groups and their information

Group	Type	Maps	Subject	University	Grade	Ects
Group 1	Individual	37	Industrial Drawing	University 1	1st Grade – Bachelor 6 Mechanical Engineering	
Group 2	Group	31	Industrial Drawing	University 1	1st Grade – Bachelor 6 Industrial Engineering	
Group 3	Individual	12	Design of Machinery	University 2	3rd Grade – Bachelor 9 Mechanical Engineering	
Group 4	Group	12	Design of Machinery	University 2	3rd Grade – Bachelor 9 Mechanical Engineering	
Group 5	Individual	51	Industrial Facilities and Energy Efficiency	University 1	1st Grade – Masters 6 Industrial Engineering	
Group 6	No Activity carried out	79	Industrial Drawing	University 1	1st Grade – Bachelor 6 Industrial Engineering	

active control groups (engaged in different activities) and have a passive control group for a thorough examination of outcomes. Additionally, the inclusion of both individual and group concept map development aimed to provide a comparative analysis of these approaches.

The rationale for selecting specific student groups considered factors such as the nature of the subject, academic level (Bachelor or Masters), and diversity in engineering fields (e.g., Mechanical and Industrial Engineering). This diversity ensured that the study captured a broad spectrum of students with varying levels of maturity and understanding within the discipline.

The decision to apply an exploratory method for some groups and not provide additional support materials was guided by the need to observe natural responses and creativity in concept map development. This approach aligns with recommendations from prior research studies (Valdivia 2021) advocating for an exploratory approach, allowing students to select their own method, and fostering a genuine representation of their understanding.

Moreover, the selection of subjects with similar ECTS, European Credit Transfer and Accumulation System, is a standard for measuring and comparing academic achievements across European higher education institutions, was aimed at ensuring a degree of comparability across groups. Similar ECTs in different subjects provided a basis for evaluating the impact of conceptual mapping in diverse academic scenarios. Overall, the chosen methodology was designed to capture a holistic view of the concept mapping process, considering individual and group dynamics, subject variations, and the exploratory nature of the method itself.

Before explaining the work procedure during the testing, the students were recruited with a general invitation to develop a new activity within the framework of the different subjects. However, a few of the participants were familiar with concept mapping, but most had not used the technique previously. Therefore, the students received a class on concept mapping in order to describe and clarify the theoretical basic concepts and the mapping

structures. About the mapping strategy to follow for the present activity, some research studies recommend the exploratory method instead of the confirmatory method (Walker & King, 2003). However, the students selected their own method to develop the concept maps.

The procedure followed considers some outstanding aspects that students have to manage inside and outside the classroom: the complexity of the working system and individual/ teamwork. Regarding the complexity analysis from the concept maps, the qualitative data is studied. Focusing on measurements, Groups 1, 3, and 5 worked individually, and the students selected the method and generated a concept map per chapter (removable and fixed joints, geometric tolerances, movement transmission, clutches, and energy efficiency). However, Groups 2 and 4 proposed a different method. In this case, they developed a concept map in groups of four or five students, considering the entire subject. They discuss and select the method with the partners. The last aspect to evaluate in this study is the perception of the students about the concept map activity and its contribution to the achievement of the skills related to each of the three subjects, done through a survey. Once the students have developed their concept maps in a previous class, they express what they have learned with the concept map method through a survey.

The assessment system proposed to the students in this subject evaluates three stages of the learning process: the learning, the conceptual organisation, and the motivation. Beginning with the first stage, the learning assessment requires analytically evaluating the concept maps using a rubric (Lucas-Molina et al., 2017). Table 2 shows the rubric and assessment tool, which consist of seven concepts to evaluate.

The evaluation process for concept maps employed a detailed rubric consisting of seven criteria, each with specific scoring methodologies. The rubric aimed to provide a comprehensive and structured assessment of the concept maps, considering various aspects of knowledge representation. Here is a more detailed description of the criteria and scoring methodology:

- Leading Concept (Score from 3 to 0, X3): This criterion assessed the clarity and prominence of the main concept in the map, with a score range of 3 strongly related to 0 not related.
- Subordinate Concepts (Score from 3 to 0, X10): Evaluated the clarity and relevance of supporting concepts. The score ranged from 3 (strong relevance) to 0 (not relevant), multiplied by a factor of 10.
- Propositions (scores from 3 to 0, X10) assessed the meaningful relationship between two concepts and the validity of these relationships. The score ranged from 3 to 0, multiplied by a factor of 10.
- Presence of Dynamic Propositions (Score from 3 to 0, X10): Evaluated the inclusion of dynamic elements such as movement, action, change of state, or dependency relationships. The score ranged from 3 to 0, multiplied by a factor of 10.
- Cross Connections and Creativity (Number of cross connections and creative connections, X10): Assessed the significance and validity of connections between different segments of the conceptual hierarchy. The score was determined by the number of cross-connections and creative connections multiplied by a factor of 10.
- Graphic Examples (Number of graphic examples, X10): Evaluated the inclusion of graphic examples supporting the described concepts. The score was determined by the number of graphic examples multiplied by a factor of 10.
- Hierarchy (Number of valid hierarchical levels, X5): Assessed the hierarchical structure of the map, ensuring subordinate concepts are more specific than the main

concept. The score was determined by the number of valid hierarchical levels multiplied by a factor of 5.

Table 2 Rubric of concept map

Criteria	Scoring	Leverage
1) Clearing	Score from 3 to 6	3
2) Subordinate	Score from 3 to 6	X
3) Connections	Score from 3 to 6	X
4) Is the relationship of meaning between two concepts indicated by the line joining them and by the color of the line?	Score from 3 to 6	10
5) Is this relationship dynamic?	Score from 3 to 6	X
6) Does the map show significant connections between the different segments of the concept map?	Number of cross connections and hierarchical relationships. They are 0	10
7) Does the map show significant and hierarchical relationships?	Number of graphic elements	X
8) Does the map present a hierarchical structure? Is each of the subordinate concepts specific and about it?	Number of valid hierarchical levels	X

The overall rubric was structured to provide a maximum score of 100, reflecting the holistic quality of the concept map. The specific weightings for each criterion were chosen to highlight their relative importance in capturing the depth and effectiveness of knowledge representation. The rubric aimed to offer both a quantitative and qualitative assessment, allowing for nuanced insights into the students' conceptual understanding and map development skills.

The main function of concept maps is to explore the knowledge and understanding level of the students in a specific topic (Watson et al., 2016). This work analyses the results from an individual and collective mapping perspective and their influence on the global mapping score.

The second analytical step is related to the measurement of the complexity level of the concept maps. In order to evaluate this aspect, four of the seven evaluable criteria have been treated. In one part, the Leading Concepts and the Subordinate Concepts, represent the information contained in the concept map. Secondly, the Cross connection between concepts (and creativity) and the Hierarchy levels show the interdependency of the concepts. The complexity ratio C has been quantified by Eq. (1):

$$C = \frac{(\text{Cross section connection quantity} \times 10) + (\text{Hierarchy level quantity} \times 5)}{7 \times 3 + 22 \times 10} \quad (1)$$

$C = (\text{Main subject score} \times 3) + (\text{Subordinate subject score} \times 10)$ (1) where Cross Connection between Concepts is the quantity of cross connections in the concept map that signifies the meaningful interconnections between different segments of the conceptual hierarchy. Multiplied by 10 in the numerator, emphasizing its importance in contributing to the overall conceptual complexity.

Hierarchy Levels is the number of valid hierarchical levels in the concept map, indicating the structured arrangement of concepts from general to specific. Multiplied by 5 in the numerator, recognizing the significance of hierarchical organization in determining conceptual complexity.

Main Subject Score and Subordinate Subject Score are the scores obtained based on the evaluation of the leading concept and subordinate concepts, respectively, using the rubric. Multiplied by 3 and 10, respectively, in the denominator, reflecting their weightings in the overall conceptual complexity calculation.

As an example, consider Eq. (2), considering a concept map, where:

- Cross connection quantity = 20 (indicating 15 meaningful connections and 5 non meaningful 0 points),
- Hierarchy level quantity = 3 (showing three valid hierarchical levels),
- Main subject score = 10 (obtained from the rubric assessment of the leading concept, being 3 of them not related with the topic so 7 scores 3 and 3 scores 0), and
- Subordinate subject score = 22 (obtained from the rubric assessment of subordinate concepts).

$$C = \frac{15 \text{ times } 10 + 3 \text{ times } 5}{7 \text{ times } 3 + 22 \text{ times } 10} = \frac{150 + 15}{21 + 220} = 0.685 \quad (2)$$

This resulting value of 0.685 indicates the conceptual complexity level, with higher values reflecting a greater degree of complexity. The complexity ratio thus provides a

numerical measure, allowing for quantitative comparisons of the conceptual complexity across different concept maps.

Finally, surveys evaluate student motivation by analysing the general proficiencies and the learning outcomes. The student assessment has a main question: “What overall assessment does the methodology use? However, four purposes make the survey easier for the students. Table 3 illustrates the parts of this as-assessment tool. The students have five levels of interest (1. Strongly disagree, 2. In dis-agreement, 3. Indecisive, 4. In agreement and 5. Strongly agree).

The first purpose evaluates the teaching methodology, learning process, student spotlight, and improvements. Therefore, this point includes four questions (e.g., does the methodology that includes the performance of conceptual maps encourage the participation of students in class?). To assess the acquisition of general skills, the survey proposes these themes from the guide for each subject. Moreover, the third purpose focuses on learning results published in the same teaching guide for the subject. Finally, the survey examines how the students have recollected and managed the information. The survey ends by asking the students the main question related to the overall assessment of the concept map methodology. As described before, this question summarises the aim of this survey, and consequently, the results of this issue will be very conclusive.

Results

The results obtained on concept map metrics are evaluated by means of the rubric. The surveys were the tool used to measure the perceptions of the students about the concept map tool.

Finally, the scores obtained by the students in the subjects under study have been analyzed. To evaluate the effect of the activity object of this paper, the use of concept maps has been added. This group did not carry out any activity and belongs to the same engineering degree and year.

Concept map metrics

One evaluator, the professor of the subject, scored each map. The maps do not grade for course credit. With the number of map metrics analysed in this study, it was important to use the explicit assessment guideline and the rubric in Table 2 to ensure consistent scoring.

Table 4 shows the results on concept map metrics evaluated through the rubric and distributed in the different sections. The rubric evaluates seven criteria, and the results are split into five groups. Table 4 presents the results of concept map metrics evaluated using the rubric outlined in Table 2. The metrics cover seven criteria for five different groups (G1 to G5) involved in the study. Mean values (MV) and standard deviations (SD) are provided for each criterion, along with the total score and complexity ratio. The complexity ratio

Table 3 Assessment survey of concept map methodology

Purposes
1: Student spotlight, evaluation of teaching methodologies and continuous improvement of teaching / learning processes

- 2: Acquisition of general skills of each subject
- 3: Learning results of each subject
- 4: Information management and collective work

Table 4 Results of concept map metrics evaluated through the rubric

	G1		G2		G3		G4		G5	
	MV	SD	MV	SD	MV	SD	MV	SD	MV	SD
1. Leading Concept [9 pts]	8,48	1,74	9,00	0,00	6	3,24	6,75	3,27	8,26	1,55
2. Subordinate Concepts [30 pts]	25,78	6,12	17,78	6,67	20,42	7,49	22,50	6,29	22,17	5,95
3. Propositions [30 pts]	9,69	7,12	8,89	7,82	18,33	9,20	16,25	8,69	12,98	6,81
4. Presence of dynamic propositions [30 pts]	6,09	7,04	0,00	0,00	8,75	7,11	10,00	7,91	6,67	7,47
5. Cross connections and creativity [X10 per element]	7,50	9,09	0,00	0,00	10,83	10,37	5,83	8,62	7,42	7,05
6. Graphic examples [X10 per element]	5,86	4,41	12,22	3,63	1,67	5,53	1,67	3,73	2,78	5,02
7. Hierarchy [X5 per element]	7,03	4,43	4,44	3,00	12,92	4,77	16,25	5,82	0,98	1,99
Total	70,44	23,75	63,17	4,92	78,92	35,68	79,25	30,45	61,26	19,91
Complexity	0,41	0,23	0,21	0,10	0,94	0,46	0,84	0,27	0,21	0,15

reflects the quantitative measure of conceptual map complexity. The table highlights variations in scores across different criteria and groups, offering insights into the strengths and weaknesses of the concept maps.

Individuality/collectivity results in map complexity: Group 2 presents a slightly lower value of conceptual map complexity; it seems that carrying out the maps collectively does not favour the creation of more complex. The maps obtained individually show a great dispersion in their results, while when they are done in a group, the results are more homogeneous and uniform.

The following Fig. 2a shows an example of a handmade concept map. The example illustrates a concept map created by students, specifically focusing on the topic of surface finish. The map includes graphical examples supporting the acquisition of concepts related to graphic representation. The figure provides a visual representation of how students incorporated graphic elements into their concept maps. This example serves as a reference for understanding the practical application of concept mapping in representing complex engineering concepts visually. In the provided example, Fig. 2b, the conceptual map exhibits limited complexity. The primary concern is the absence of cross-correlators. If we consider the colour system and capitalization, which separate hierarchies, we end up with four levels, resulting in a score of 20. In the denominator, there are three main subjects, which contribute a value of 9, and eight subordinate subjects, which, when multiplied by 10, contribute 80. As a result, the complexity score is calculated to be 0.225. As can be seen in Eq. (3):

$$C = \frac{0 \text{ times } 10 + 4 \text{ times } 5}{3 \text{ times } 3 + 8 \text{ times } 10} = \frac{0 + 20}{9 + 80} = 0.225 \quad (3)$$

This analysis suggests that while the conceptual map offers a structured representation of the subject matter, its complexity is somewhat limited. The lack of cross-correlators and the reliance on colour and capitalization to denote hierarchy might not fully encapsulate the intricate relationships between concepts. In addition, it would be necessary to interpret and evaluate in the complexity index the appearance of graphic examples, which in this case has not been taken into account.

Progressive complex conceptual learning: the results show that C complexity ratio is slightly superior as the student advances in their academic level. On average, third-grade students produce more complex maps than first-year master's students.

Students perception through the surveys

The results obtained from the survey conducted in class show the degree of student satisfaction with the concept mapping activity. Table 5 outlines the results obtained from surveys conducted for various purposes defined in the study. Mean values and standard deviations are presented for each purpose (G1 to G5), reflecting student perceptions of the concept mapping activity. The purposes include student spotlight and evaluation of teaching methodologies (Purpose 1), acquisition of general skills (Purpose 2), learning results for each subject (Purpose 3), and information management and collective work (Purpose 4). The table provides a nuanced understanding of how students perceive the concept mapping activity across different evaluation criteria.

The results presented in Table 5 show that the concept mapping activity i to the students. Group 1 is considered more positive than Groups 3 and 4, with the lowest satisfaction value (3,16 from 5, versus 3,78), but positive for all the studied groups.

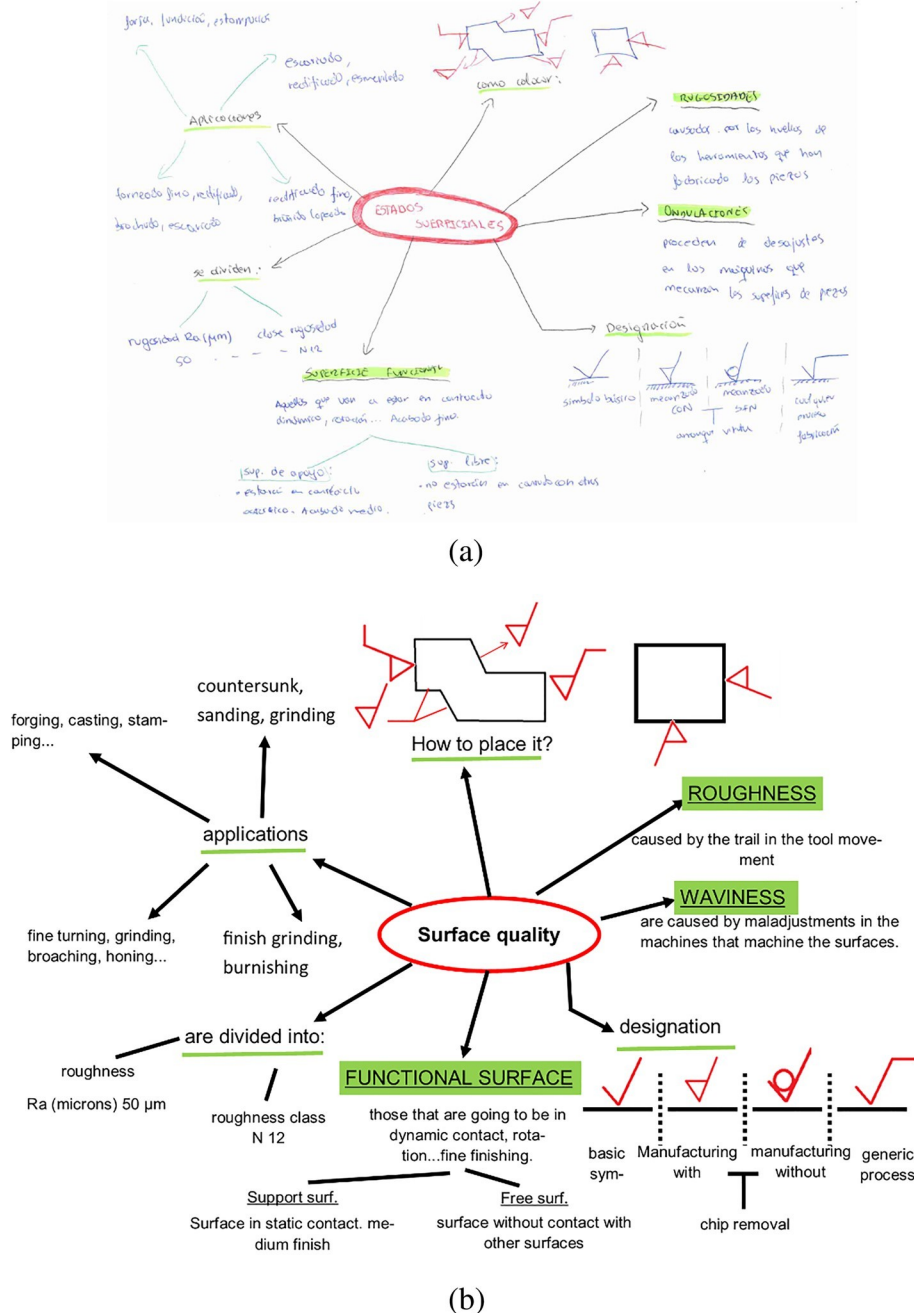


Fig. 2 Example of a concept map for the surface finish: **a** original map, **b** digitalized version

Table 5 Results of the survey according to previously defined purposes

	G1		G2		G3&4	G5		
	Mean value	Std dev	Mean value	Std dev	Mean value	Std dev	Mean value	Std dev
Purpose 1	3,68	0,84	3,49	0,89	3,24	1,03	3,48	0,39
Purpose 2	3,72	0,75	3,43	0,79	3,00	1,00	3,57	0,54
Purpose 3	3,78	0,80	3,35	0,85	3,06	0,90	3,25	0,59
Purpose 4	3,92	0,79	3,62	0,81	3,35	1,17	4,27	0,74
Total	3,78	0,80	3,47	0,84	3,16	1,03	3,64	0,66

The survey results regarding the perceived usefulness of concept maps for different purposes reveal some interesting discrepancies among student groups. Understanding these variations requires an exploration of potential reasons rooted in educational approaches and contextual factors.

Students' scores through practical and theoretical tests

Table 6 shows the results obtained from the different tests of knowledge conducted in class. An additional group of control is analysed, Group 6, this group did not perform any activity related to the conceptual map experiment. The table presents the scores obtained from different tests (Test 1, Test 2, and Test 3) conducted in class for groups G1 to G6. Mean values and standard deviations are provided, considering 10 as the maximum score. Test 1 evaluates theoretical concepts; Test 2 assesses practical knowledge; and Test 3 covers overall knowledge. Additionally, an additional control group (G6) that did not participate in any concept mapping activity is included for comparative analysis. The table offers insights into the impact of the concept mapping activity on students' performance in different knowledge assessments.

The results in Table 6 show that the concept mapping activity led to better results in all the tests performed. Being especially significant, the improvement obtained in the results of Test 1 is considered as a test of theoretical concepts. Group 1 led this score with a high value (9,83 from 10, versus 7,8 and finally, the group that not perform any activity related to this work had a score of 5,93). Finally, it is possible to confirm that the concept maps are useful to improve the teaching and learning of new concepts. The overall valuation of the knowledge improves from the previously obtained 6,15 for both groups involved in this experimental work (Group 2: 6,98 and Group 1: 7,19).

The qualitative analysis of map complexity involved an in-depth examination of various findings derived from the concept map analysis. Here are some discussions on the assessment of complexity:

Group 2, which created maps collectively, presented a slightly lower value of conceptual map complexity. This suggests that collaborative map creation may not necessarily lead to more complex maps. The analysis considered the overall concentration of complexity scores.

Maps created individually showed a greater dispersion in results, while those done in groups exhibited more homogeneous and uniform outcomes.

Table 6 Scores obtained on the previously defined tests (considering 10 as the maximum score)

	Group 1 (es)- Completed samples		Group 2 (es)-Completed samples		Group 3 (es)- Completed samples		Group 4 (es)- Completed samples		Group 5 (es)- Completed samples		Group 6 (es)- No activity carried out	
	Mean value	Std	Mean value	Std	Mean value	Std	Mean value	Std	Mean value	Std	Mean value	Std
Test 1	83	0 51	80	0 82	25	0 91	05	13	23	05	93	19
Test 2	94	0 36	7 61	0 89	8	0	7	1	7 12	1 23	5 98	1 04
Test 3	53	1 04	7 30	1 63	-	-	-	-	8 90	1 97	5 59	1 84
Total	19	1 88	6 98	1 51	05	91	05	13	8 10	0 26	6 15	1 01
	7	0	6	1	8	0	7	1	8	1	6	1

Discussion

The analysis of cognitive complexity and concept maps offers a unique perspective into the cognitive complexity of students' conceptual maps (Novak & Cañas, 2008). This approach transcends mere content assessment, enabling educators to assess the complexity of the mental structure's students construct. The integration of qualitative analyses of cognitive complexity into the curriculum can assist instructors in aligning teaching methods with students' cognitive development. This approach promotes a more nuanced understanding of individual learning processes, a finding that aligns with previous research on cognitive development (Conceição et al., 2017).

The discovery that conceptual map complexity does not necessarily correlate with academic progression challenges traditional assumptions. This suggests that students' ability to handle complex engineering concepts may be influenced by factors beyond their academic level. These finding echoes earlier studies that advocate for a departure from a linear understanding of cognitive development and a consideration of diverse factors influencing students' conceptualization skills. Tailored interventions addressing individual needs can be designed, recognizing that complexity is not solely a product of academic advancement (Krieglstein et al., 2022).

The perception of usefulness and valued purposes underscores the tool's utility beyond individual learning. This aligns with the evolving landscape of collaborative work in engineering fields, a trend noted in previous research (Izci & Akkoc, 2024; Krieglstein et al., 2022). Educational practices should leverage concept maps as collaborative tools, emphasizing skills in information management and collective problem-solving. The integration of these aspects into the curriculum prepares students for collaborative work environments prevalent in industrial and engineering settings.

The comparative results and improvements underscore the tangible impact of this pedagogical approach. The comparative analysis with a control group underscores the effectiveness of concept maps in enhancing learning outcomes, a finding consistent with earlier studies. Institutions should consider incorporating concept mapping activities as a standard practice within the curriculum. The demonstrated improvement suggests that such activities contribute significantly to knowledge retention and comprehension.

The analysis of individual vs. group work suggests that individualized approaches better capture the hierarchical structures of students' minds. While group work remains valuable, instructors should recognize the unique strengths of individual concept mapping. Although when it comes to concept mapping, individual construction can sometimes lead to limited perspectives, while collaborative work fosters creativity (Anastasiou et al., 2024), given the limitations of this work, collaborative work may have faced coordination and consensus challenges as it performed worse in all areas of assessment. Balancing individual and collaborative assessments ensure a comprehensive evaluation of students' conceptualization abilities.

In synthesizing educational practices, it is recommended to embrace a variety of instructional strategies that accommodate diverse learning preferences and cognitive styles. Concept maps can coexist with other methods to create a comprehensive learning environment. Designing curricula that holistically address both individual and collective learning needs aligns with previous research advocating for diverse instructional strategies. The integration of collaborative elements prepares students for the teamwork inherent in engineering practice. It is recognized that educational practices need to continuously adapt.

Concept maps, while a valuable evaluation tool, have certain limitations and challenges that need to be considered. These include dependency on skill and experience, time constraints, lack of additional support materials, influence of the subject nature, varied interpretation of rubric criteria, evaluator bias, interpretation of complexity ratio, survey subjectivity, and generalization across subjects. These limitations echo those identified in previous research and underscore the need for careful consideration when implementing concept maps as an evaluation tool.

Despite its potential benefits, the use of concept mapping as an evaluation tool could be seen as unfair to some students. This is particularly true for those who may be new to the technique or who may struggle with the specific skills it requires (Plotz, 2020). On the other hand, concept mapping can be an incredibly effective tool for teaching and learning. It encourages students to visualize relationships between concepts, which can enhance understanding and retention. It also promotes higher order thinking and helps students develop skills in critical analysis and synthesis of information.

Therefore, while concept mapping may have its limitations as an evaluation tool, its strengths in promoting learning and understanding make it a valuable technique in the educational process. It's crucial, however, to provide adequate support and training for students to ensure they can effectively use and benefit from this method. This aligns with previous research advocating for comprehensive support when implementing new instructional techniques.

Conclusions

Based on the results presented, the authors identify several implications of the application of concept maps as a teaching and learning improvement tool. In conclusion, this study contributes to the existing body of knowledge on concept mapping by shedding light on its effectiveness in assessing the organisation and representation of knowledge in educational settings, especially for engineering education at different levels and with group and individual testing. Through a comprehensive analysis of various scoring methods and their implications, this research adds nuance to our understanding of how concept maps can be integrated into teaching practices. Building on previous studies, which have highlighted the importance of concept maps in promoting student understanding and facilitating collaborative learning (Cook-Sather, 2002; Novak & Cañas, 2008), this study extends the discourse, providing insights into the practical considerations involved in concept map scoring. Our findings emphasize the need for educators and researchers to carefully select scoring methods that align with their goals.

Through the calculation of the C complexity ratio, the qualitative measurement of students' conceptual complexity understanding levels can be conducted. Concept maps serve as representations of individuals' cognitive structures, allowing for analysis of both their content and complexity. The results indicate that higher grades are associated with greater complexity in the represented maps, particularly evident in the complexity index, though not consistently reflected in the total score. Interestingly, some undergraduate students have produced more comprehensive concept maps, resulting in higher total scores than some master's students. Students perceive information management and collective work as the most valued purposes of concept maps. Comparative analysis between the subjects where the activity was implemented and the control group (Group 6) reveals

significant improvements, with an average increase in grades of more than 1 point out of a maximum possible 10 points.

Furthermore, the study's results underscore notable disparities among the various groups based on their approach to completing concept mapping tasks. Specifically, Group 1, engaging in individual completion, consistently outperformed those undertaking group-based completion (Groups 2 and 4). While these findings shed light on the efficacy of concept mapping, they also emphasize the importance of considering additional variables, such as individual learning preferences and activity design, when implementing this methodology in educational environments. Exploring alternative concept mapping methods, like students empty in a blank concept map structure with instructor-provided word banks (Pierre-Antoine et al., 2014) or constructing maps using pre-defined concepts and linking phrases (Franklin et al., 2016), could offer intriguing alternatives to further investigate.

The utilization of maps as a supplementary activity provides a convenient method for assessing subjects' levels of comprehension. Individual maps consistently yield superior overall results, as evidenced by both rubric-based scores and student perceptions collected through surveys. This preference for individual work or assessment arises from the tool's capacity to reflect the unique mental hierarchies of individuals on specific topics. However, it seems to contradict previous assertions that emphasize the benefits of collaborative concept mapping (Boxtel et al., 2002). This discrepancy may be influenced by the paper's limitations, such as focusing on engineering students and concept mapping within class time.

To further enhance the concept mapping assessment process, future research could explore several avenues. One promising approach involves addressing the reliance on skills and expertise by providing training or support to students in concept mapping. This could empower them to develop more intricate propositions and connections. Additionally, considering the impact of subject nature on concept mapping assessment could lead to tailored approaches, acknowledging that certain subjects lend themselves more naturally to visual representation. Investigating the generalizability of results across subjects and educational contexts and refining the assessment process accordingly would contribute to a nuanced understanding of the effectiveness of concept mapping. Overall, integrating qualitative and quantitative measures, engaging multiple raters, and considering subject-specific factors could lead to a comprehensive and unbiased evaluation of concept maps in educational settings.

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Declarations

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Conflict of interest The authors declare that they have no conflict of interest.

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